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SUBSECTION 4.5 INSTRUMENTATION AND CONTROL

4.5.1. INSTALLATION

4.5.1.1 Control Panels

Control panels shall be installed in accordance with **Part 4. Subsection 4.4.**

4.5.1.2 Instruments

Locally mounted instruments shall be installed & supported in a neat and workmanlike manner and arranged to facilitate maintenance. The Bidder shall furnish and install junction boxes as required and shall wire all instruments to junction boxes. Instruments shall be grouped wherever possible without unduly extending the length of sensing lines, particularly the sensing lines for low pressure and differential pressure measurement. Final grouping and arrangement shall be as approved by the Owner.

Grouped instruments shall be mounted in local mounted instrument panels which may be shop or field fabricated by the Bidder.

The Bidder shall furnish and erect all necessary stands, brackets and supports for instruments not mounted in panels. Instruments and panels shall not be mounted on or supported by grating, handrails, hangers, cable trays, or piping and duct hangers nor on rotating or vibrating equipment unless they form an integral part of such equipment and the manufacturer has demonstrated by shop tests or field experience the suitability of the installation.

Instruments such as pressure gauges and dial thermometers shall not be mounted directly on piping if the piping is susceptible to vibration or if the location of the tapping point or well would make it difficult to read a directly mounted indicator. Pressure test points shall be installed in an accessible location or means of access shall be furnished, and erected by the Bidder.

Isolating valves shall be furnished and installed in the sensing lines close to all instruments where the main line shut off valves are in an inaccessible location.

Pressure taps in horizontal liquid process piping shall be located on the side of the piping.

Pressure taps in horizontal process piping for air, and gases, and condenser and low pressure heater steam shall be located on the top of the piping.

Instrument installation details showing the above details for each instrument shall be submitted for Owner approval.

4.5.1.3 Thermowells

The Bidder shall install all thermo wells at Site, except that welded type thermo wells in shop fabricated piping shall be installed and stress-relieved in the shop.

The Bidder shall submit his installation and stress relieving procedures for all welded type thermo wells whether shop or site installed to the Owner for review.

Thermo wells in horizontal pipes shall be installed in the side of pipes where possible. All thermo wells shall be located in an accessible location or means of access shall be furnished and erected by the Bidder. The location of thermo wells shall permit the easy insertion and withdrawal of temperature indicators, switches and elements.

The Bidder shall ensure that all thermo wells have sufficient lagging extension so that the temperature elements or indicators can be installed, inspected or replaced without interfering with the insulation.

All thermo wells shall be furnished with screwed metal plugs attached to the wells by chains. The Bidder shall ensure that all plugs remain in the thermo wells, to prevent the ingress of foreign material and liquid into the well, until the elements or indicators are installed in the wells.

4.5.2. PIPING AND TUBING

Complete instrumentation piping and tubing shall be furnished. This shall include but not be limited to the following:

Sensing and sampling piping, tubing, fittings, siphons, and valves etc. between main line shut-off valves and the instrumentation devices or panels.

Instrument air supply piping, tubing, fittings, cocks etc. between instrument air header shut-off valves and the instrumentation devices.

Instrument air signal piping, tubing, fittings, cocks etc. between instrumentation devices and panels.

Capillary tubing for filled system temperature measurement and control.

Requirements for the supply and installation of piping and tubing shall be as follows:

Unless otherwise specified, sensing and sampling piping and tubing shall conform to the classification of piping material and design condition for the main piping to which it is connected in accordance with **Part 4 Section 3, M- 001**.

Instrument air supply and signal piping and tubing shall conform with the requirements of **Part 4 Section3, M-001**.

Piping and tubing lines shall be properly spaced, pitched and supported. They shall be located so as to be protected from damage and deflection. Piping and tubing connected to any place subject to movement due to thermal expansion shall be installed with provision to allow sufficient movement of the tapping point without breaking or stressing the piping tubing and connections or distorting the tapping points. Sampling lines shall have, provision for movement longitudinally due to thermal expansion. Fittings shall be so located that the line movement does not cause them to come into contact with the pipe supports, etc.

Channel sections or other suitable means shall be used where necessary for proper protection and for support as required. Where single or multiple runs of copper or capillary tubing run through a floor they shall be protected for a distance of approximately 2 meters above the floor in conduit or other suitable enclosure.

Capillary tubing shall not be kinked or bent sharply. An surplus length of tube shall be neatly coiled and hung concealed behind the instrument or in another inconspicuous location.

Bends in piping and tubing shall be made with a proper pipe bending machine or tool and shall have sufficient radius to avoid kinking or wrinkling of the pipe or tube wall.

Leak testing requirements shall include the following:

All piping and tubing shall be tested to identify leaks. All leaks shall be repaired and the piping and tubing retested. The final test shall demonstrate complete freedom from leaks.

Sensing and sampling lines shall be hydrostatically tested at pressures suitable for the process to which they are connected where the process pressure is above 7 bar.

Sensing and sampling lines for process pressures below 7 bar and all air lines shall be tested with air or nitrogen at a pressure of not less than 7 bar. Soapsuds, or the commercial equivalent, chlorine or chloride free fluid, shall be applied to each joint to detect leakage. During the test, all welded and brazed joints shall be hammered to loosen any scale. When all detected leaks have been repaired the lines shall be pressurized, sealed off, and the source of air disconnected. There shall be no appreciable pressure drop after 8 hours.

After the piping and tubing has been proved to be leak free, it shall be blown through with clean air to remove any foreign material or liquid.

4.5.3. FLOW NOZZLES AND ORIFICE PLATES

Installation of the laboratory tested feed water flow nozzle condensate flow nozzle and straightening vane flow section; all other flow sections and all

other flow measuring elements shall be in accordance with the latest edition of ASME “Fluid Meters, Their Theory and Application”.

The Bidder shall take micrometer measurements of the internal diameter of the pipe, upstream and downstream of the flow sections and elements. All measurements shall be witnessed by the Owner’s representative and entered on the certified data sheets for the flow sections and elements. The certified data sheets shall be submitted to the Owner for review.

Installation of flow sections and elements shall be supervised by the Manufacturer’s representative or a suitably qualified supervisor selected by the Bidder and approved by the Owner’s representative. Installation shall be witnessed by the Owner’s representative.

The Bidder shall install welding neck type orifice flanges for all orifice plates complete with flange taps, jacking screws, bolts, nuts and gaskets, suitable for the design pressures and temperatures and in accordance with the **Part 4 Section 3, M-001**.

4.5.4. CALIBRATION AND PREPARATION FOR USE

The Bidder shall prepare all instruments and control devices for use. Protective material, straps, braces and stops shall be removed. Parts removed for shipping shall be installed. Any instruments or devices, or the parts or glasses thereof, broken or damaged during shipping shall be replaced. Charts and chart printing devices shall be installed in recorders.

The Bidder shall ensure that all pressure instruments are properly corrected for static liquid head and shall make readjustments if necessary.

The calibration of all instruments shall be checked and calibration records prepared for the Owner’s use.

The Bidder shall cooperate fully with the Owner to ensure that all control systems and loops are correctly interconnected.

4.5.5. PRECOMMISSIONING TESTS

4.5.5.1 Instrument Tests

Each gauge, transmitter, flow meter, measuring element, analyzer, actuator and other instrument shall be tested as soon as practicable after receipt on site but in all cases prior to installation. Each instrument is to be calibrated in the upscale and downscale directions and if necessary adjusted until the accuracy conforms to the limits stated by the manufacturer.

The approval of the owner shall be obtained in writing before any non-standard modifications or adjustments are made;

Each test shall be carried out at 0%, 25%, 50%, 75%, 100% upscale and downscale of, the instrument range and also at intermediate ranges when required by the Owner.

Upon completion, calibration certificates signed by the Bidder shall be handed to the Owner. Records of abortive tests shall also be handed to the Owner.

Following the test, each instrument shall either be immediately installed in the plant or shall be labeled and carefully packed and stored in a dust free environment.

4.5.5.2 Hardware Testing on Site

It will not be assumed that equipment leaving the manufacturer's works in good order will arrive on Site in equally good order.

Therefore, each module of every system shall be inspected on site, and shall be demonstrated to be functional before the system is confirmed to be ready for a full system test. Particularly close attention shall be paid to any item showing signs of imperfect packing, mishandling, or other potential damage or faults.

Tests covering the testing of individual loops, all binary and analogue input output signals from the plant to the system input terminals in the Local or Central Control Room shall be undertaken.

Transmitter output signals equivalent to 50 and 100 percent of the instrument range shall be generated to check the response of all other instruments, modules and control valves in the loop.

Manual operation of control valves from Local or Central Control Rooms shall be checked as shall valve positioned indicators.

The auto/manual transfer scheme shall be checked to ensure 'bump less' transfer between auto and manual.

4.5.5.3 Complete Systems Tests on Site

The function of the Site test is to prove that the system functions to specification in the environmental conditions of site and over the complete range of the power supply variations and that it has been transported and erected satisfactorily, and to prove that the system is sufficiently reliable to justify connecting it to the plant.

Before the formal system tests are allowed to commence, it is the Bidders responsibility to ensure that sufficient checks and tests have been performed (all to be documented) to ensure that the system is in a fit state to be fully tested. This activity to be referred to as preparation for site acceptance test is to be shown on the Bidders bar chart schedule.

Site tests on the I & C system shall be based on the Works Test programmed and shall utilize primary (pressure, temperature, DP, etc) injection testing at

each transmitter, to completely verify connections, cabling, and other installation work as well as the I & C cubicles, panels and desk indications. Further, the command signaling from auto/manual controls on desks and panels shall be checked through to each actuator drive system.

Prior to connecting incoming plant signals to the plant interface modules of the DCS, RIO, PLC, etc. if used, all functional tests of the Works System Test shall be repeated and completed to the satisfaction of the Owner. The functional tests shall be performed again when all the plant connections have been made.

In the event of low frequency of system or equipment problems occurring at Works and Site, the Owner may waive the requirement to repeat the 250 hour test, and require instead a 100 hour test.

The tests are to demonstrate that the system is capable of handling a maximum normal load while maintaining an adequate response to both operators and plant, in accordance with the Technical Requirements of this Specification.

4.5.5.4 Maintenance Up to Acceptance

The Bidder shall list all modules, which are sent off-site for repair. Wherever possible the Bidder shall demonstrate repair and maintenance of items needing such service to the station staff by utilizing the test and calibration equipment provided under this contract.

At the time of takeover by the Owner, the Bidder shall replace or refurbish all such test and calibration equipment to the satisfaction of the Owner. This equipment, at takeover, is to be in proper working order and in a clean undamaged state.

4.5.6. COMMISSIONING TESTS

4.5.6.1 General

The Bidder shall perform commissioning tests for all instrumentation and control devices, loops, sub-systems and systems.

The Bidder shall cooperate fully with the Owner during the commissioning of instrumentation and control loops, sub systems and systems.

Commissioning tests for instrumentation and controls shall be carried out independently of and or together with the mechanical systems they control as applicable.

The Bidder shall perform fine tuning of control systems.

4.5.6.2 Schedule

Commissioning tests shall start after erection is essentially complete and continue until the Owner issues the Taking Over Certificate as specified in **Part 2, Clauses 54 and 55** except that by mutual agreement between the

Owner and the Bidder minor fine tuning of the control systems may continue after the Taking Over Certificate has been issued.

Commissioning tests shall, as far as is practicable, be scheduled as follows:

Functional tests on devices and loops as soon as instrumentation & control erection is complete.

Functional tests on subsystems as soon as the applicable devices & loops have been tested.

Functional tests on systems as soon as the applicable subsystems have been tested and before the associated mechanical equipment is ready for operation.

Performance tests on systems in conjunction with the associated mechanical equipment.

Reliability Run on the distributed digital control system as soon as the Unit is available to be operated in a manner and for a length of time suitable for the Reliability Run.

Fine tuning of control systems whenever mechanical equipment and processes are available for controlled operation.

4.5.6.3 Test on Devices

Tests on devices (in addition to calibration) shall include but not be limited to the following:

Control Valves

Control valves and actuators shall be checked for correct stroking, range and direction. All accessories such as solenoid valves, positioners, limit switches; position transmitters etc. shall be checked for correct functioning. Failure - position of the valve on loss of air or electrical power shall be demonstrated to be as indicated on the Valve Data Sheet.

Switches

Process connected switches shall be checked for correct functioning of the operating mechanism. Contact closing and opening shall be verified and demonstrated to be as indicated on the instrument data sheet. (e.g. closed for alarm condition etc.)

Recorders

Recorders shall be checked for correct functioning of chart drives (including speed) inking or printing mechanism, auxiliary switches, doors and locks.

Printers, Keyboards and VDU Displays

Printers, keyboards and VDU displays shall be functionally tested and exercised using the manufacturer's standard test routines and procedures.

Push Buttons and Control Switches

Push buttons and control switches shall be checked for correct functioning of operating mechanisms.

Annunciators

Annunciators shall be checked in accordance with **Part 4, Subsection 4.4** if applicable and for correct functioning in accordance with the requirements of the relevant Technical Specification.

4.5.6.4 Tests on Loops

The Bidder shall verify that the configuration of each loop is as indicated on the applicable loop diagram.

Pneumatic loops shall be checked for correctness of supply, input signal and output signal air tubing connections.

Where possible, tests on devices shall be coordinated with tests on loops. e.g. switch action with annunciator or control system response; set point or manual control operation with valve stroking etc.

4.5.6.5 Tests on Subsystem & Systems

Wherever possible subsystems and systems shall be functionally tested using simulated inputs and with all display, printing and recording devices, keyboards, alarm annunciators and final control elements operating prior to tests in conjunction with associated mechanical equipment.

Full functional and performance tests shall be carried out in conjunction with the associated mechanical equipment.

A 90 days Availability Run shall be carried out on the distributed digital control system in accordance with **Part 4 Section 3, I-005**.

Fine tuning of the response of control systems shall be carried out by initiating changes and minor upsets to mechanical equipment and processes and recording the response of the control system in bringing the equipment or process to the new operating point or back to the original operating point as applicable. Adjustments to the control characteristics and repeated tests shall be made until optimum control is achieved.

(Last page of Subsection 4.5)